



UNIVERSITY OF THE PUNJAB
B.S. 4 Years Program : Fifth Semester – Fall 2021
Paper: Quantum Mechanics-I

Course Code: PHY-305

Roll No.
Time: 3 Hrs. Marks: 60

(10x3=30)

Q.1. Give short answers to the following questions.

- i. How do you define term "Wave Function" in quantum mechanics. Write down its physical significance.
- ii. Show that Hamiltonian operator and momentum operator commutes with each other.
- iii. Explain any three postulates of quantum mechanics. 2022, 23, ** ✓
- iv. Show that is an eigen function of an operator and determine eigen value if 2022 ✓
- v. Write down physical significance of position momentum uncertainty. 2022 ✓
- vi. What is zero-point energy? Calculate zero-point energy of one-dimensional harmonic oscillator.
- vii. Consider two states
Find the value of 'a' so that and are orthogonal. ✓
- viii. State correspondence principle ✓
- ix. Show that
- x. Define ladder operator in quantum mechanics. Give their importance in any physical phenomenon. ✓

Answer the following questions in detail.

Q2. (a). Derive equation of continuity and explain its physical significance in hydrodynamics system. ✓

(b). If is normalized wave function. Find the expectation value of where $0 < x \leq 10$.
(5+5)

Q3. A particle of mass m and total energy $E > V_0$ strikes a potential step from left where
Calculate reflection and transmission coefficient. Also show that $R+T=1$ (10)

Q4. (a) Obtain eigen value of . ✓

(b) Can z-component of angular momentum and azimuthal angle can be measured with precision at the same time. ✓
(6+4)

M.P

The variable quantity describing De-Broglie wave associated with moving particles is called wave function. The wave function has no physical significance since it is a variable quantity. It may be expressed as:

$$\psi(x,t) = a + ib.$$

Q#1

where, "a" and "b" are real. Now,

$$\psi^*(x,t) \psi(x,t) = (a - ib)(a + ib)$$

$$\Rightarrow |\psi(x,t)|^2 = a^2 + b^2$$

Thus, the product of $\psi^*(x,t)$ and $\psi(x,t)$ is real and positive if $\psi(x,t) \neq 0$. Its positive square root is denoted by $|\psi(x,t)|$ and the quantity $|\psi(x,t)|^2$ is called probability density.

Q. 2.2:- What is physical significance of wave function?

Answer.-The wave function $\psi(x, t)$ has no physical significance directly. Its modulus square $|\psi(x, t)|^2$ has physical meaning, which gives probability density i.e. the quantity $|\psi(x, t)|^2 dx$ probability of finding the particle at time 't' in a distance dx. The total probability of finding the system somewhere in space is always unity. Note that the wave functions $\psi(x, t)$ and $e^{i\alpha}\psi(x, t)$ represent same state, α being a real constant.

Q. 2.3:- Some wave functions are given below:

Q#2:- To show that Hamiltonian operator $\hat{H}$ and momentum operator $\hat{P}$ commute with each other.

Q#2

$$[H, P] = 0$$

where $[A, B] = AB - BA$ is the commutator.

Now, $[H, P] = HP - PH$ $\because H = \frac{P^2}{2m} + V(x)$

" Put values of H & P " $\because P = -i\hbar \frac{\partial}{\partial x}$

$$= \left[\frac{P^2}{2m} + V(x) \cdot (-i\hbar \frac{\partial}{\partial x}) \right] - \left[(-i\hbar \frac{\partial}{\partial x}) \cdot \left(\frac{P^2}{2m} + V(x) \right) \right]$$

By expanding,

$$[H, P] = \left[\cancel{-i\hbar \left(\frac{P^2}{2m} \cdot \frac{\partial}{\partial x} \right)} - \cancel{i\hbar V(x) \frac{\partial}{\partial x}} + \cancel{i\hbar \frac{\partial}{\partial x} \frac{P^2}{2m}} + i\hbar \frac{\partial}{\partial x} V(x) \right]$$

$$[H, P] = 0$$

This demonstrates that the $[H, P] = 0$ H and P commute with each other.

Write down three postulates of Quantum Mechanics? ③

Ans: The formalism of quantum mechanics is based on a number of postulates. These postulates are in turn based on a wide range of experimental observations.

According to classical mechanics, the state of a particle is specified at any time t , by two fundamental dynamical variables;

1) the position $\hat{x}(t)$

2) the momentum $\hat{p}(t)$

Q#3

But the quantum mechanical counterparts to these ideas are specified by these postulates to know about the system information at microscopic level.

There are some basic postulates of quantum mechanics

1- With every physical system, we associate a mathematical object denoted as ψ it is called the state-vector / State-functions / wave-function.

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MHA-3

L.O - 2018

It is a function of space and time co-ordinates and it contains all possible information about the system.

The properties of ψ are given below:

- ψ is continuous
- ψ is differentiable
- ψ and its derivative are finite and continuous
- ψ is single valued
- ψ is square integrable $\langle \psi, \psi \rangle < \infty$

2- With every physical quantity, we associate a linear hermitian operator which when acts on ψ gives us the value of physical quantity.

The operators are generally represented by a Capital letter with a hat on it, such as $\hat{A}$

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③

L.O - 2018

MHA-3

3- When operator $\hat{A}$ acts on ψ , there are two possibilities:

(i) ψ is the eigen-function of $\hat{A}$ and the eigen value λ is the required value of physical quantity

(ii) If ψ is not an eigen function of $\hat{A}$, then to measure the physical quantity we take expectation value of $\hat{A}$ in the state ψ gives as

$$\langle \hat{A} \rangle = \frac{\langle \psi | \hat{A} | \psi \rangle}{\langle \psi | \psi \rangle}$$

:- The expectation value is the average of repeated measurement

~~(Q#iv)~~ To show that a function $\psi(x)$ is an eigen function of an operator $\hat{A}$, we need to demonstrate that:

$$\hat{A}\psi(x) = a\psi(x)$$

where "a" is an eigen value.

Q#4

$$\therefore \hat{A} = \frac{d}{dx}, \quad \psi(x) = e^{ix}$$

Now,

$$\begin{aligned} A\psi(x) &= \frac{d}{dx}(e^{ix}) \\ &= i \underbrace{e^{ix}} \\ &= i\psi(x) \end{aligned}$$

$$\text{So, } \boxed{a = i}$$

$$\text{eigen function} = \psi(x)$$

$$\text{eigen value} = \boxed{i = a}$$

$$\text{operator} = \hat{A}$$

where $h = h/2\pi = 1.054 \times 10^{-34} \text{ J}\cdot\text{s}$

Physical significance of uncertainty principle: The uncertainty principle leads to following conclusions:

- (i) - If momentum p_x of a particle is determined with accuracy at some instant i.e. $\Delta p_x = 0$, then at same instant, the uncertainty in position Δx becomes infinite.
- (ii) - If position coordinate of a particle is determined with accuracy at some instant $\Delta x = 0$, then at same instant, the uncertainty in momentum Δp_x becomes infinite.

Q#5

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ILMI Quantum Mechanics (Volume 1)

Thus if an experiment is performed to measure position x accurately, then its momentum will become uncertain and vice versa. Both position and momentum can be measured within certain limits of accuracy specified by uncertainty relation.

(iii) - For a particle of mass m moving with velocity ' v ', uncertainty becomes;

$$\Delta x m \Delta v_x \geq \frac{h}{4\pi} \Rightarrow \Delta x \Delta v_x \geq \frac{h}{4\pi m}$$

For heavy bodies $\frac{h}{4\pi m} \approx 0$, the uncertainties vanish and position and momentum can be measured simultaneously. This is limiting case of classical mechanics. Thus, classical mechanics is true for heavy bodies and uncertainties are characteristic of quantum mechanics.

Energy-Time

For a one-dimensional harmonic oscillator, the zero-point energy (E_0) can be calculated using the following formula:

Q#6

$$E_0 = (1/2)\hbar\omega$$

where:

- $\hbar$ is the reduced Planck constant ($\hbar = h/2\pi$)
- ω is the angular frequency ($\omega = \sqrt{k/m}$)

Substitute the expression for ω :

$$E_0 = (1/2)\hbar\sqrt{k/m}$$

Now, let's use the following values:

- $\hbar = 1.0545718 \times 10^{-34}$ J s (reduced Planck constant)
- $k = 1$ N/m (spring constant)
- $m = 1$ kg (mass)

Substitute values into the equation:

$$\begin{aligned} E_0 &= (1/2) \times 1.0545718 \times 10^{-34} \\ &\quad \sqrt{(1/1)} \\ &= 5.2739619 \times 10^{-35} \text{ J} \end{aligned}$$

This is the zero-point energy in one dimension for the given harmonic oscillator.

Hint:-

Energy eigen values:- Energy eigen values of harmonic oscillator are given by,

$$E_n = \hbar\omega \left(n + \frac{1}{2} \right), \quad n = 0, 1, 2, 3, \dots$$

where $\hbar$ is Plank's constant, ω is angular frequency and n is quantum number having values given above.

Zero point energy:- The minimum energy of harmonic oscillator is called zero point energy or ground state energy.

For $n = 0$, $E_0 = \frac{\hbar\omega}{2}$ is called zero point energy or ground state energy.

20(18, 2018)

Wave function for ground state:- Expression for ground state of harmonic oscillator is given by,

$$\psi_0 = A_0 e^{-\frac{m\omega x^2}{2\hbar}}$$

Put $a = \frac{m\omega}{\hbar}$, then

$$\psi_0 = A_0 e^{-\frac{ax^2}{2}}$$

By applying normalization condition, we find $A_0 = \left(\frac{a}{\pi} \right)^{\frac{1}{4}}$. So normalized wave function for ground state of harmonic oscillator is,

$$\psi_0 = \left(\frac{a}{\pi} \right)^{\frac{1}{4}} e^{-\frac{ax^2}{2}}$$

~(Q#7)~ we consider two state,

$$\psi_1(x) = e^{ix}$$

$$\psi_2(x) = e^{ax}$$

Q#7

where,

$$\int \psi_1^*(x) \psi_2(x) \cdot dx = 0$$

" $\psi_1^*(x)$ is the complex conjugate of $\psi_1(x)$ ",
put values,

$$\int e^{(-ix)} \cdot e^{ax} \cdot dx = 0$$

$$\int e^{(ax-ix)} \cdot dx = 0$$

$$\because ax-ix = (a-i)x \\ = i(a-1)x$$

$$\int e^{i(a-1)x} \cdot dx = 0 \quad \text{integral must be zero,}$$

$$a-1=0$$

$$a=1$$

"The value of 'a' that makes $\psi_1(x)$ & $\psi_2(x)$ orthogonal."

But $\int_{-\infty}^{\infty} \psi^2 dx = 1$

$A = \int_{-\infty}^{\infty} \psi^2 dx$

2.5: The correspondence principle:—When Bohr tried to work out the structure of atoms based on nuclear model of electrons orbiting the nucleus at centre, he faced some problems. Bohr model was based on discrete energy levels but a classical atom could be characterized by continuous spectrum. Bohr was puzzled to see that there is one set of rules for quantum systems and a different set of rules for classical systems. He resolved his problem with the help of correspondence principle, which can be stated as: "Quantum theory must agree with classical theory in the limit of large quantum numbers."

Another way of stating this principle is,

"Quantum theory must agree with classical theory in the limit in which classical theory is known to agree with experiment."

The correspondence principle shows that we do not need to draw a line between classical and quantum mechanics. In order to understand this principle, let us apply quantization of energy to motion of simple pendulum: "

From experience we know that a simple pendulum can oscillate with any reasonable kinetic energy and not only with discrete values. As friction causes the amplitude of pendulum to decrease, it appears that energy of pendulum is dissipated in continuous way but not in jumps or "quanta". Since ' h ' is so small, the jumps may be there but too small to detect and hence we cannot dismiss energy quantization. In fact in this case classical theory works well, so there is no need to apply quantum theory. This example leads us to a conclusion that classical theory is a limiting case of quantum theory in which $h \rightarrow 0$.

Just as Newton's 2nd law is basis of most elementary principles of classical mechanics, motion of quantum mechanical particles can be dealt by Schrodinger wave equation. However Schrodinger wave equation looks very different from Newton's 2nd law, but its outcomes reduces to Newton's 2nd law for large systems. Hence in classical physics, we obtain solutions by Newton's laws and in quantum physics by Schrodinger equation.

2.6: Complementarity's principle:—The principles of interference and diffraction

Q#8

iv) Briefly explain the physical significance of ladder operators in Quantum Mechanics.

Ans: The operators " L_+ " and " L_- " are known as ladder operators because these operators operate on eigen function raise or lower eigen value step wise where as

$$\hat{L}_+ = \hat{L}_x + i\hat{L}_y$$

$$\hat{L}_- = \hat{L}_x - i\hat{L}_y$$

Q#10

In Quantum Mechanics the raising operator is called creation operator bcz it adds a quantum in the eigen value and lowering operator is called

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annihilation operator.

The significance is that the ladder operators acting on a state do not affect the eigenvalue of the quadratic operator " L^2 ".

The basic commutator algebra is:

$$[L_i, L_j] = \epsilon_{ijk} L_k$$

where ϵ_{ijk} are the totally antisymmetric.

LONG

Q#1 Pg #281

Q#2 (Spring 2023)

Q#3 (ilmi Pg #235)